

On the magnification of gravitational lens images near cusps

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Abstract. On the base of the gravitational lens equation obtained by Schneider & Weiss the magnification of images near a cusp is investigated. Using the symmetrical polynomials on the roots of the polynomial of the third degree we slightly generalize the Schneider & Weiss statement on the magnification near different solutions of the gravitational lens equation. The analytical expressions for magnifications of different images near the cusp are presented.

Key words: gravitational lensing

1. Introduction

It is well known that the mapping of two-dimensional surfaces into a plane gives only two types of stable singularities: folds and cusps (pleats). There are also similar singularities of caustics in gravitational lens optics. Schneider & Weiss (1986; 1992) studied the gravitational lens mapping near the cusps. Mandzhos (1993) investigated the mutual coherence by solving analytically the gravitational lens equation near the cusp. In turn, we study the magnification near the cusp and obtain a useful analytical expression for it.

2. Basic equations

We recall the basic equations from Schneider & Weiss (1992) before considering their gravitational lens equation near the cusp. As is shown in Schneider, Ehlers & Falco (1992), hereafter SEF, the gravitational lens equation may be written in the following form: let the distance between an observer and a source be D_s , the distance between an observer and the gravitational lens be D_d , and D_{ds} be the distance between the gravitational lens and a source. If we suppose a small angle of deflection then we have the following simple expression for the lens equation (SEF)

$$\eta = D_s \xi / D_d + D_{ds} \beta(\xi),$$

where the vectors η, ξ define the coordinates in source plane and in the lens plane respectively (SEF),

$$\beta(\xi) = 4G/c^2 \int \rho(\mathbf{R}) (\xi - \mathbf{R}) / |\xi - \mathbf{R}|^2 d\mathbf{X}d\mathbf{Y},$$

where $\mathbf{R} = \{X, Y\}$ is the point vector in the lens plane, $\rho(\mathbf{R})$ is the surface mass density of the gravitational lens. We introduce the following variables (SEF)

$$x = \xi / R_0, \quad y = D_s \eta / (R_0 D_d),$$

where $R_0 = \sqrt{2r_g D_d D_{ds} / D_s}$ is the Einstein - Chwolson radius (see Historical Remarks in SEF). We also introduce the following notation for the scaled (SEF) angle

$$\alpha = \beta D_{ds} D_d / (D_s R_0).$$

In the modelling of gravitational lenses, the surface mass density is normalized with the critical surface mass density (Wamsganss 1990)

$$\rho_{cr} = \frac{c^2 D_s}{4\pi G D_d D_{ds}}.$$

For typical lensing situations the critical surface mass density is of the order of $\rho_{cr} = 10^4 M_\odot pc^{-2}$ (Wamsganss 1990). Therefore, if we define the scaled surface mass density by the following expression (SEF)

$$\sigma = \rho / \rho_{cr},$$

then we have the expression for the angle α

$$\alpha(\mathbf{x}) = \int \sigma(\mathbf{x}') (\mathbf{x} - \mathbf{x}') / |\mathbf{x} - \mathbf{x}'|^2 d^2 \mathbf{x}'.$$

As Schneider (1985) showed, we may introduce the scalar potential ψ , such, that

$$\alpha(\mathbf{x}) = \nabla \psi(\mathbf{x}),$$

where

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$$\psi(\mathbf{x}) = \int \sigma(\mathbf{x}') \ln |\mathbf{x} - \mathbf{x}'| d^2x'.$$

Then (SEF)

$$\mathbf{y} = \mathbf{x} - \alpha(\mathbf{x}) \quad (1)$$

If we introduce (as in SEF)

$$\phi(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})^2/2 - \psi(\mathbf{x}),$$

we can write the lens equation in the form (Schneider 1985)

$$\nabla \phi(\mathbf{x}, \mathbf{y}) = 0.$$

It is easy to see that mapping $\mathbf{x} \mapsto \mathbf{y}$ is a Lagrange mapping (Arnold 1979), since that is gradient mapping (Arnold 1983). Really, if we consider the function

$$S = \mathbf{x}^2/2 - \psi(\mathbf{x}),$$

then

$$\mathbf{y} = \nabla S.$$

Singularities of Lagrange's mappings are described in Arnold's paper (1972), and in Arnold's review (1983), and their bifurcations are described in Arnold's paper (1976).

Equation (1) defined the mapping of points on the lens plane into points on the source plane. Using the Jacobian matrix we define the local mapping (SEF):

$$\delta \mathbf{y} = A \delta \mathbf{x},$$

$$A = \begin{pmatrix} \partial y_1 / \partial x_1 & \partial y_1 / \partial x_2 \\ \partial y_2 / \partial x_1 & \partial y_2 / \partial x_2 \end{pmatrix} = \begin{pmatrix} 1 - \psi_{11} & -\psi_{12} \\ -\psi_{21} & 1 - \psi_{22} \end{pmatrix}, \quad (2)$$

where

$$\psi_{ij} = \frac{\partial^2 \psi}{\partial x_i \partial x_j} (i, j = 1, 2). \quad (3)$$

Let us consider lenses which are systems of point masses. Out-sides the points where the masses are located, we have the following equality (SEF)

$$\psi_{22} = -\psi_{11}, \quad (4)$$

since the potential obeys the Laplace equation.

The magnification of an image at $\mathbf{x}$ is

$$\mu(\mathbf{x}) = 1 / \det A(\mathbf{x}),$$

since the distortion of the transformation is determined by the Jacobian matrix. The singular points of the mapping are characterized by

$$\det A(\mathbf{x}) = 0$$

(where the mapping is not one-to-one). The set is formed by so-called critical curves (SEF).

Since $\det A(\gamma^{(c)}) = 0$ (at points on the critical curve), it is possible to find coordinates with $\phi_{11} \neq 0, \phi_{12} = \phi_{22} = 0$. As shown in Schneider & Weiss (1992), the lens equation near the cusp can be represented as

$$\begin{aligned} y_1 &= cx_1 + \frac{b}{2}x_2^2, \\ y_2 &= bx_1x_2 + ax_2^3, \end{aligned} \quad (5)$$

where $a = \frac{1}{6}\phi_{2222}, b = \phi_{122}, c = \phi_{11}$ and $c \neq 0, b \neq 0, 2ac - b^2 \neq 0$. It was shown in SEF that additional terms do not affect the local properties of the mapping. It is possible to see that by the direct comparison of terms (for example the term with ϕ_{112} is smaller than the term with ϕ_{11}) or using Newton's polygon or Bruno's truncating rules (see for example Bruno 1989). Similar to Schneider & Weiss (1992), we introduce the notations

$$\hat{x}_1 = \frac{c}{b}x_1, \quad \hat{x}_2 = x_2, \quad \hat{y}_1 = y_1/b, \quad \hat{y}_2 = \frac{c}{b^2}y_2, \quad (6)$$

then we have

$$\begin{aligned} \hat{y}_1 &= \hat{x}_1 + \frac{1}{2}\hat{x}_2^2, \\ \hat{y}_2 &= \hat{x}_1\hat{x}_2 + s\hat{x}_2^3, \end{aligned}$$

where $s = ac/b^2$. We also introduce $S = 1 - 2s, \tilde{y}_1 = 2\hat{y}_1/3S, \tilde{y}_2 = \hat{y}_2/S$; then we have (Schneider & Weiss 1992)

$$\hat{x}_1 = \frac{3S}{2}\tilde{y}_1 - \frac{1}{2}\hat{x}_2^2, \quad (7)$$

$$\hat{x}_2^3 - 3\tilde{y}_1\hat{x}_2 + 2\tilde{y}_2 = 0. \quad (8)$$

3. Statement on the magnifications of images near cusps

We recall that (Schneider & Weiss 1992)

$$\hat{\mu} = (\det \hat{A})^{-1}, \quad \hat{\mu} = b^2 \mu, \quad (9)$$

$$\det A = b^2 \det \hat{A} = b^2[\hat{x}_1 + (3s - 1)\hat{x}_2^2]. \quad (10)$$

Consider the magnifications for different images of one point inside the cusp. We show below that the following equality is valid for all sources inside the cusp

$$\hat{\mu}^{(1)} + \hat{\mu}^{(2)} + \hat{\mu}^{(3)} = 0. \quad (11)$$

It is clear from Eq.(11) that we have the statement of Schneider & Weiss (1992) that

$$|\hat{\mu}^{(1)}| = |\hat{\mu}^{(2)} + \hat{\mu}^{(3)}|. \quad (12)$$

It should be mentioned that Eq. (11) follows from Schneider and Weiss Eq.(2.3) (Eq. (12) of our paper) plus their comment on the parities after their Eq.(2.33b). We use the following expression for magnifications

$$\mu^{(i)} = \frac{1}{\hat{x}_1^{(i)} + (3s - 1)(\hat{x}_2^{(i)})^2} \quad (13)$$

or using Eq.(7)

$$\mu^{(i)} = \frac{2}{S[\tilde{y}_1 - (\hat{x}_2^{(i)})^2]} \quad (14)$$

Therefore it is necessary to prove that

$$\frac{1}{\tilde{y}_1 - (\hat{x}_2^{(1)})^2} + \frac{1}{\tilde{y}_1 - (\hat{x}_2^{(2)})^2} + \frac{1}{\tilde{y}_1 - (\hat{x}_2^{(3)})^2} = 0 \quad (15)$$

or that

$$3\tilde{y}_1^2 - 2\tilde{y}_1[(\hat{x}_2^{(1)})^2 + (\hat{x}_2^{(2)})^2 + (\hat{x}_2^{(3)})^2] + [(\hat{x}_2^{(1)}\hat{x}_2^{(2)})^2 + (\hat{x}_2^{(1)}\hat{x}_2^{(3)})^2 + (\hat{x}_2^{(2)}\hat{x}_2^{(3)})^2] = 0.$$

Vieta's theorem applied to Eq.(8) yields

$$\begin{aligned} \hat{x}_2^{(1)} + \hat{x}_2^{(2)} + \hat{x}_2^{(3)} &= 0, \\ \hat{x}_2^{(1)}\hat{x}_2^{(2)} + \hat{x}_2^{(1)}\hat{x}_2^{(3)} + \hat{x}_2^{(2)}\hat{x}_2^{(3)} &= -3\tilde{y}_1. \end{aligned}$$

If we express the symmetric power polynomials in terms of symmetric elementary polynomials we have

$$(\hat{x}_2^{(1)})^2 + (\hat{x}_2^{(2)})^2 + (\hat{x}_2^{(3)})^2 = 6\tilde{y}_1, \quad (17)$$

$$(\hat{x}_2^{(1)}\hat{x}_2^{(2)})^2 + (\hat{x}_2^{(1)}\hat{x}_2^{(3)})^2 + (\hat{x}_2^{(2)}\hat{x}_2^{(3)})^2 = 9\tilde{y}_1^2, \quad (18)$$

Thus we obtain (16). Similarly we have

$$q_1 = \hat{\mu}^{(1)}\hat{\mu}^{(2)}\hat{\mu}^{(3)} = \frac{2}{(3S)^3(\tilde{y}_1^3 - \tilde{y}_2^2)}, \quad (19)$$

$$p_1 = \hat{\mu}^{(1)}\hat{\mu}^{(2)} + \hat{\mu}^{(1)}\hat{\mu}^{(3)} + \hat{\mu}^{(2)}\hat{\mu}^{(3)} = \frac{-3\tilde{y}_1}{(3S)^2(\tilde{y}_1^3 - \tilde{y}_2^2)}. \quad (20)$$

Therefore we get an equation of third degree for the magnification of different images of a point inside the cusp

$$\tilde{\mu}^3 + \tilde{p}_1\tilde{\mu} + \tilde{q}_1 = 0, \quad (21)$$

where $\tilde{\mu} = \hat{\mu}/S$, $\tilde{p}_1 = p_1(3S)^2$, $\tilde{q}_1 = q_1(3S)^3$. If $\tilde{y}_1^3 > \tilde{y}_2^2$ then the discriminant for the pure cubic equation

$$D = \left(\frac{\tilde{p}_1}{3}\right)^3 + \left(\frac{\tilde{q}_1}{2}\right)^2 = \frac{-\tilde{y}_2^2}{(\tilde{y}_1^3 - \tilde{y}_2^2)^3}$$

is negative and we have three solutions of the Eq.(21) (see for example, Bronstein & Semendjajew 1980). Namely, we have

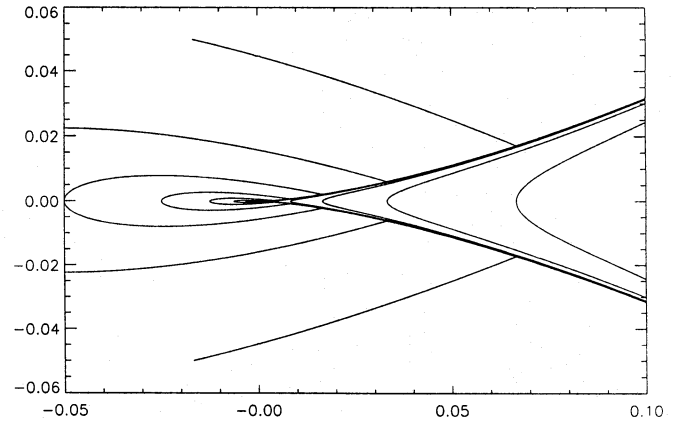


Fig. 1. Magnification contours around a cusp for the sum of absolute values of magnifications of all images. The contour levels are $10 * 2^i, -1 \leq i \leq 14$

for a point inside the cusp region

$$\tilde{\mu}^{(1)} = -\frac{2}{3}\sqrt{\frac{\tilde{y}_1}{\tilde{y}_1^3 - \tilde{y}_2^2}} \cos \left\{ \frac{\cos^{-1} \left[\sqrt{\frac{\tilde{y}_1^3 - \tilde{y}_2^2}{\tilde{y}_1^3}} \right]}{3} \right\}, \quad (22)$$

$$\tilde{\mu}^{(2)} = -\frac{2}{3}\sqrt{\frac{\tilde{y}_1}{\tilde{y}_1^3 - \tilde{y}_2^2}} \cos \left\{ \frac{\cos^{-1} \left[\sqrt{\frac{\tilde{y}_1^3 - \tilde{y}_2^2}{\tilde{y}_1^3}} \right]}{3} + \frac{2\pi}{3} \right\}, \quad (23)$$

$$\tilde{\mu}^{(3)} = -\frac{2}{3}\sqrt{\frac{\tilde{y}_1}{\tilde{y}_1^3 - \tilde{y}_2^2}} \cos \left\{ \frac{\cos^{-1} \left[\sqrt{\frac{\tilde{y}_1^3 - \tilde{y}_2^2}{\tilde{y}_1^3}} \right]}{3} + \frac{4\pi}{3} \right\} \quad (24)$$

and we have only one solution for a point outside the cusp region

$$\tilde{\mu}^{(1)} = \frac{\sqrt[3]{\sqrt{\tilde{y}_2^2 - \tilde{y}_1^3} + \tilde{y}_2} + \sqrt[3]{\sqrt{\tilde{y}_2^2 - \tilde{y}_1^3} - \tilde{y}_2}}{\sqrt{\tilde{y}_2^2 - \tilde{y}_1^3}}. \quad (25)$$

It is possible to calculate the magnifications of the images from the direct solution of the gravitational lens equation near a cusp. Namely, we solve Eq.(8) and use Eqs.(7),(9),(10). After that we also obtain the expressions for magnifications.

4. Discussion

In Fig. 1 we present the contours for the sum of absolute values of magnifications of images near the cusp. Note that the algebraic sum of the values is equal to zero and that there are three real solutions of the gravitational lens equation in the cusp region. The contours are plotted by using expressions (22-25). It is possible to compare Fig. 1 with the similar figure from Schneider & Weiss (1992) that was obtained using the ray - shooting method (see, for example, Wamsganss 1990). Figure 1 shows clearly the cusp type singularity.

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References

- Arnold V.I., 1972, *Funct. Anal. and Appl.* 6, 619 (in Russian)
Arnold V.I., 1976, *Comm. Pure Appl. Math.* 29, 557
Arnold V.I., 1979, *Mathematical Methods of Classical Mechanics*, Nauka, Moscow (in Russian)
Arnold V.I., 1983, *Uspehi Mat. Nauk.* 38, N 2, 77 (in Russian)
Bronstein I.N., Semendjajew K.A., 1980, *Taschenbuch der Mathematik*, Verlag Harri Deutsch Thun und Frankfurt/Main
Bruno A.D. 1989, *Local Methods in Nonlinear Differential Equations*, Springer Verlag, Berlin
Mandzhos A.V. 1993, *A Zh* (in press)
Schneider P., 1985, *A&A* 143, 413
Schneider P., Ehlers J. and Falco E. 1992, *Gravitational Lenses*, Springer, New York
Schneider P. and Weiss A., 1986, *A&A* 164, 237
Schneider P. and Weiss A., 1992, *A&A* 260, 1
Wamsganss J. 1990, *Gravitational Microlensing*, Dissertation der Fakultät für Physik der Ludwig - Maximilians - Universität, München